

Akshat G

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Summary

AI/ML engineer and MedTech startup co-founder specializing in production ML systems, computer vision, LLMs, and GPU-accelerated inference. Experience building real-time perception pipelines, leading technical teams, and shipping end-to-end AI systems from research to deployment.

Education

Manipal Institute of Technology

B.Tech (Honours), Computer Science and Engineering — AI Specialization

Bengaluru, India

Jul 2023 – May 2027

- **CGPA:** 9.56/10

Professional Experience

Shell India Markets Private Limited

Intern, Multiphysics AI and Product Innovation

Bengaluru, India

Jun 2026 – Present

- Deployed a client-side AI assistant for defect detection and automated report generation, cutting report turnaround from ~40 hours to 2 hours; kept human-in-the-loop review for high-value expert validation.
- Designed a modular inference framework decoupling feature extraction from classification, accelerating multi-backbone experimentation.
- Developed HiME (Hierarchical Multilabel Estimator), a novel architecture for hierarchical gear failure classification (27 classes, ~1.05M label combinations) — top performer among 16 ensembled agents and the architectural foundation for 10 of them; achieved 100% hierarchy consistency and 91% root-to-leaf accuracy.

Sensera – [\[DOI\]](#)

Co-Founder & ML Engineer

Bengaluru, India

Feb 2026 – Jun 2026

Core ML infrastructure and proprietary medical AI assets acquired following product development.

- Co-founded an AI MedTech startup; shipped a full-stack Personal Health Record platform with 50+ daily active users.
- Built an async document pipeline decoupling OCR/NLP inference across 10 parallel GPU workers, achieving sub-3-second document processing and extraction.
- Implemented semantic search over medical records (pgvector, row-level security) and a self-hosted LLM pipeline for summarization and structured extraction — sub-second, user-isolated retrieval with no dependence on external inference APIs.

CEAM — Center of Excellence in Autonomous Mobility

Perception Lead, Project MAVERIC – [\[DOI\]](#)

Bengaluru, India

Aug 2025 – Aug 2026

- Led perception for MAVERIC, a fully autonomous go-kart, building the complete real-time sensor fusion stack — custom OS, vision pipeline, and object tracking.
- Led a 16-member computer vision team; built perception pipelines in optimized C++ (Eigen) and NumPy — 56 Hz semantic segmentation, 16 FPS object detection.
- Developed a custom LiDAR-camera fusion pipeline for 3D scene understanding and real-time path planning.

Publications [\[Google Scholar\]](#) [\[ORCID\]](#)

2 indexed conferences and 5 accepted journals in data-centric AI, semi-supervised learning, and few-shot learning — *Scientific Reports*, *SAGE Digital Health*, Springer CCIS; peer reviewer, *IEEE Transactions on Medical Imaging*.

Technical Skills

- **Programming:** Python, C/C++, Java, TypeScript/JavaScript, SQL, Bash
- **ML Frameworks & Tools:** PyTorch, TensorFlow, HuggingFace Transformers, Sentence-Transformers, TensorRT, ONNX Runtime, OpenCV, scikit-learn
- **AI/ML Specialization:** LLMs, RAG, Dense Retrieval, Semantic Search, Vector Embeddings, Computer Vision, Transfer Learning, Fine-tuning
- **Core Expertise:** Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI
- **Distributed Systems & Cloud:** Docker, Modal (Serverless GPU), Supabase, Google Cloud (OAuth, Drive API), AWS, Git
- **Databases:** PostgreSQL (Schema Design, RLS, Migrations, Indexing), pgvector (HNSW ANN)
- **Frontend & Backend:** React Native (Expo SDK 54), React, FastAPI, Pydantic